EEE341 Lab 5:Antenna Measurement

1 Objective

To study radiation pattern, gain, realized gain and impedance characteristics of a dipole antenna and two-element dipole arrays using EZNEC.

2 Introduction

An antenna is a device used to radiate electromagnetic energy into free space. An antenna can also be used to receive electromagnetic energy from free space. Antennas are usually designed to work in a certain frequency range within which they have to serve following functions:

1. Direct energy efficiently into one or more directions.
2. Match the impedance of the system to that of free space.
3. Provide the desired polarization.

The dipole antennas simulated in this lab experiment are designed to be resonant at a frequency of approximately 300MHz corresponding to a wavelength of 1 meter. (Note that in EEE 341, we have assumed that the speed of light in vacuum is 3×10^8 m/s. A more precise value is 2.99793×10^8 m/s. So a wavelength of 1 meter corresponds more precisely to a frequency of 299.793 MHz which is the default frequency in the EZNEC software.)

2.1 Radiation Pattern and Half Power Beamwidth

Antennas are designed to direct energy into specific directions. The angular variation in the radiated power is described by the radiation pattern or beam pattern. The angular extent of the main lobe where the radiated power drops to one-half its maximum level is known as the Half Power Beam Width (HPBW).

Radiation pattern can be measured by placing a standard gain antenna at a fixed location turning the antenna under test to various angles. Such measurements are done in an antenna range which may be open or in a special room lined with absorbing materials.

2.2 Far-Field Region

Far-field region is the region beyond at certain distance from the phase center of the antenna where in the shape of the antenna radiation pattern does not change with distance. The phase center of the antenna is the point from which it appears that the spherical waves emanated from the antenna appear to originate.

2.3 Impedance

The antenna acts as a load Z_A connected at the end of the waveguiding system. If Z_A is not matched to the system impedance, then there will be reflections and not all the power sent by the RF source will be radiated.

2.4 Antenna Gain

Antenna gain (G) is a parameter that describes the variations of the radiation intensity as a function of direction. For a lossless isotropic source the gain is 1. Gain (also called "matched gain") is the product of directivity and radiation efficiency. Gain increases if the antenna is more directive and decreases with the losses in the antenna. "Realized gain" also includes the effect of impedance and polarization mismatch. As impedance and/or polarization mismatch losses increase, realized gain decreases.

2.5 Antenna Arrays

Very often desired radiation characteristics cannot be achieved using a single antenna. This limitation can often be overcome by using an antenna array. An antenna array is an electrical and geometric arrangement of discrete radiating elements (which are usually but not always identical). For arrays of identical elements, the total far-zone electric field can be written as

$$\overrightarrow{E} = \overrightarrow{E_\circ} \times \mathrm{AF} \tag{2.1}$$

where $\vec{E_\circ}$ is the electric field of a single element at a reference point, and AF is the array factor which depends on the geometric and electrical relationships among elements. (Here we are assuming that the mutual coupling among the elements is negligible.)

For a two-element array with the elements placed along the z-axis and fed in phase with equal amplitudes, the array factor is given by

$$\mathrm{AF}(\theta) = 2\cos\left(\frac{\pi d}{\lambda} cos\theta\right) \tag{2.2}$$

where d is the spacing between the two elements and λ is the wavelength in free space. For a two-element array with the elements placed along the x-axis and fed with equal amplitudes but a phase difference of ξ, the array factor is given by

$$\mathrm{AF}(\theta, \phi) = 2\cos\left(\frac{\pi d}{\lambda}\sin\theta\cos\phi + \frac{\xi}{2}\right) \tag{2.3}$$

2.6 Azimuth and Elevation

Assume that a coordinate system has been set up such that the +z-axis points toward the zenith. In this situation, we can assume that the angle ϕ is the azimuth angle, and that the elevation angle is given by $90° - \theta$. An azimuth pattern is generated by setting $\theta = 90°$ and varying ϕ. An elevation pattern is generated by fixing ϕ, e.g.,$\phi = 0$ and varying θ.

2.7 Friis Equation

The Friis equation gives the power received by one antenna from another antenna some distance away that is transmitting a given amount of power. It is given by

$$\frac{P_r}{P_t} = G_t G_r \left(\frac{\lambda}{4\pi R} \right)^2 \tag{2.4}$$

where P_r is the received power, P_t is the transmitted power, G_r is the matched gain of the receive antenna, G_t is the matched gain of the transmit antenna, λ is the wavelength in free space, and R is distance between transmit and receive antennas.

3 Procedure

To start you should download and install the EZNEC program available from `http://eznec.com/`, and view the recommended tutorial materials. Open the example file **Dipole1.EZ**, and save it to your working directory under another name such as **HWD.EZ**. (Hint: use HWD.EZ file attached on Lab5 page in lieu of Dipole1.EZ.)

3.1 Plotting the SWR

Under **Setups** → **Frequency Sweep...** setup a frequency sweep as shown in Fig.3.1. To start the simulation press the **Freq Swp** tab that has appeared on the lower left of the control center. Once the simulation is completed, press the **SWR** button to view your results which should appear as illustrated in Fig.3.2.

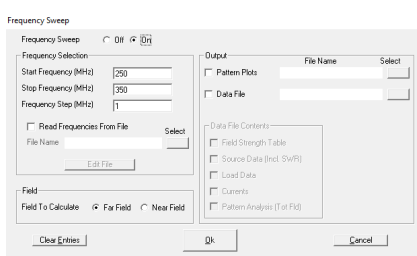

Figure 3.1: Frequency sweep setup used in simulation of half-wave dipole.

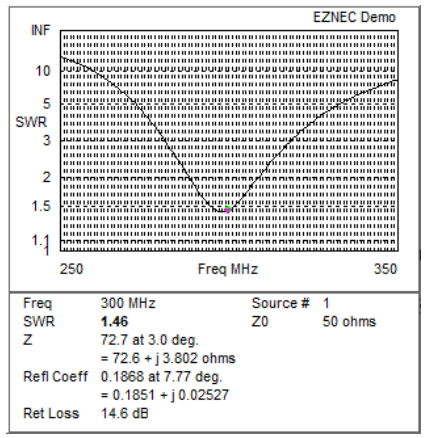

Figure 3.2: SWR as a function of frequency for half-wave dipole.

3.2 Plotting the Radiation Pattern

On the main control panel, go back into **Setups** and turn off the frequency sweep. Make sure the frequency shown in the control panel is 299.793MHz. Push the arrow next to **Plot Type** on the main control pain to toggle between Azimuth, Elevation and 3 Dimensional. Press the **FF Plot** button to produce the plot. Make sure your result for the 3 Dimensional plot looks like that in Fig.3.3.

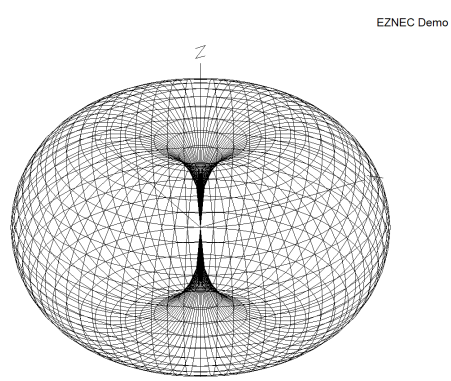

Figure 3.3: 3D pattern of half-wave dipole.

3.3 Two-Stacked-Element Array

To create a two-element array, push the arrow next to **Wires** on the main control pain to bring up the **Wires** panel. Under the **Wire** menu, select **Copy Wires...** and copy the existing wire with a z-offset of 0.5 meters (one-half wavelength). Be sure that the **Copy sources**, **loads**, **TL stubs** box are checked. The number of segments on each wire should be set as 11 segments. Generate an elevation plot of the far field for this geometry.

3.4 Cardioid Array

To create a cardioid array, place two half-wavelength dipoles 0.25 meters (one-quarter wavelength) apart. Set up the source on the second dipole so that it phase lags the first by 90 degrees. The **Wires** and **Sources** panels should be set up as shown in Fig.3.4. Generate an azimuth plot of the far field for this geometry.

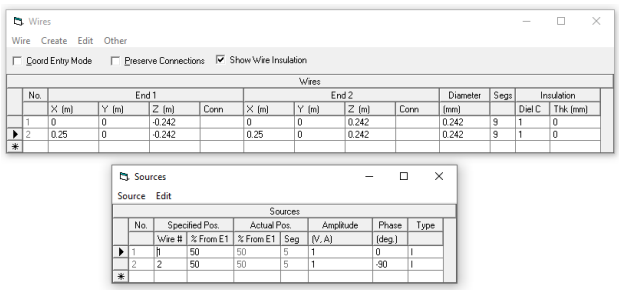

Figure 3.4: Setup of Wires and Sources Panels for cardioid array.

3.5 Communications Link Simulation

In this exercise, you will simulate the link between two half-wavelength dipoles separated by 20 meters. Arrange two half-wavelength dipoles so that they are separated by 20 meters along the x-axis. Only one of the dipoles will be excited by a source; the other will be terminated in a 50 ohm load. The **Wires**, **Sources** and **Loads** panels should be set up as shown in Fig.3.5. Determine the power delivered to the load of the second antenna (receiver) from the source of the first antenna (transmitter). To determine the power supplied by the source and the power delivered to the load, press the **Src Dat** and **Load Dat** buttons respectively.

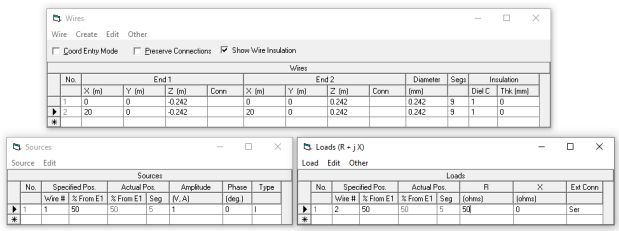

Figure 3.5: Setup of Wires, Sources and Loads Panels for communications link simulation.

4 Lab Report

Utilizing report template to complete the assigned tasks. Compile your work into a single PDF file, and submit it using the link provided in Canvas. Be sure to include a **header with course title**, **your last name** and **Lab**. For instance, EEE341_ your last name_ Lab5.pdf.